



Experimental studies on VIV performance and countermeasures for twin-box girder bridges with various slot width ratios

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ABSTRACT

The slot width ratio of a twin-box girder plays an important role in the aerodynamic performance of the corresponding bridge. Therefore, it is necessary to investigate the VIV performance of a twin-box girder with various slot width ratios and evaluate the effectiveness of present VIV countermeasures considering the influence of different slot width ratio. Through a series of wind tunnel tests, the present study firstly investigated the VIV performance of a twin-box girder suspension bridge with six representative slot width ratios ranging from 0% to 100%. Then, four countermeasures, including increasing structural damping ratio, installing grid plates, wind barriers and guide vanes, were checked and comprehensively evaluated with the aim of identifying the most effective VIV control measure. The experimental results showed that the VIV performance of a twin-box girder with 0% slot width ratio (a single box girder) is superior to those with central slot. The maximum normalized heaving VIV amplitude occurs at the slot width ratio of 60% for the 1:60 geometrical scale model (around 0.093) and 80% for the 1:30 scale model (around 0.1), respectively. The significant difference of the torsional VIV responses between the two models with different geometrical scale is found at the slot width ratio of 20%. The comparison of VIV countermeasures for the 20% and 60% slot width ratios suggested that increasing damping ratio is helpful for reducing VIV responses, while installing guide vanes may have adverse effects. The application of grid plates and wind barriers can have positive effects in suppressing the heaving VIV responses, while the effectiveness of grid plates is more obvious with the decrease of grid porosity. Furthermore, the application of the grid plates with 0% grid porosity can perfectly enable the VIV performance of the twin-box girder bridge to meet the requirements of both Sperling indicator and Wind-resistance specification of China.

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1. Introduction

Vortex-induced vibration (VIV) is a major challenge for flexible structures subjected to wind actions in terms of structural safety and serviceability, such as long-span bridges, tall buildings, chimneys, power transmission towers, etc. It is well

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known that when the frequencies of vortex shedding are close to the bending or pitching natural frequencies of long-span bridges, it may induce large-amplitude vibrations, resulting in the discomfort of the drivers and long-term fatigue damages in bridge structures (Ehsan and Scanlan, 1990; Lee et al., 1997). Most importantly, long-span bridges are often subjected to vortex-induced vibrations at low wind velocities even below 10 m/s, famous examples of which are Rio-Niterói Bridge (Battista and Pfeil, 2000), Storebælt Suspension Bridge (Larsen et al., 2000), Trans-Tokyo Bay Bridge (Fujino and Yoshitaka, 2002), and Xihoumen Suspension Bridge (Ge et al., 2011). As a complicate bluff body, the main girder of a long-span bridge cannot always achieve the perfect aerodynamic optimization in the wind-resistance design. Therefore, it is necessary to accurately predict the VIV performance and effectively mitigate possible vortex-induced vibrations by implementing practical countermeasures in the wind-resistance design of long-span bridges.

As a new type of bridge girder, twin-box girders can significantly improve the aerodynamic instability performance of long-span bridges, so become increasingly popular in the construction of super long-span cable-supported bridges, for instance, Xihoumen Bridge with the main span of 1650 m (China) and Gwangyang Bridge with the main span of 1545 m (Korea). Although more and more attentions have been paid to the VIV performance of long-span bridges with twin-box girders in recent years (Larsen et al., 2008; Ge et al., 2011; Li et al., 2011; Kwok et al., 2012; Laima et al., 2013; Yang et al., 2014), achieving the fully comprehension of VIV performance for a long-span bridge with a twin-box girder is still a challenge for bridge engineers. For example, most of these researches are confined to one certain slot width ratio. Moreover, it has been proven that the flutter instability performance of twin-box girders with various slot width ratios is different from each other (Yang et al., 2009; Yang et al., 2015; Chen et al., 2014). However, the role of slot width ratio played in VIV performance of twin-box girders has not been completely clarified, so a systematical investigation is required. In addition, previous studies have suggested that the sectional models with larger geometrical scale ratios associated with higher Reynolds numbers are likely to predict the vortex shedding and VIV responses of twin-box girder bridges more accurately (Larsen et al., 2008; Zhang et al., 2008). To further understand the influence of Reynolds numbers on the VIV performance, VIV tests of twin-box girder bridges with various slot width ratios should be conducted under lower and higher Reynolds number conditions.

When the displacements of vortex-induced vibrations are large, it is necessary to apply effective control measures on the bridge structures. Generally, VIV countermeasures can be divided into three different categories, i.e., aerodynamic, structural, and mechanical countermeasures (Fujino and Yoshida, 2002; Chen et al., 2013). One of the commonly used aerodynamic countermeasures is the application of additional components on a bridge girder, such as wind fairings, guide vanes, and wind barriers. On the other hand, the modification of air flow around the main girder by attaching or adjusting the parameters of some non-structural components with particular shape (e.g. railings, maintenance rail, crash barriers) at particular location of the girder has also proven to be useful (Zhou et al., 2015). VIV countermeasures of twin-box girders have been extensively studied in the last decade. Through a series of wind tunnel tests, Battista and Pfeil (2000) explored the effectiveness of a system of tuned-mass dampers (TMDs) on decreasing the intense VIV responses observed on RioNiterói Bridge with a steel twin-box bridge deck. Li and Ge (2008) found that the vortex control effect of guide vane is closely correlated to the height, width and angle of guide vane itself, as well as the slot width. Ge et al. (2011) revealed that the method of increasing structural damping ratio and adjustable wind barriers has better VIV control effect of the Xihoumen Bridge if compared to the guide vanes and grid plates. Yang et al. (2014) evaluated the effectiveness of three countermeasures (e.g. guide vanes, grid plates and wind barriers) on mitigating VIV of three different types of twin-box girders bridges. Furthermore, the study of Laima et al. (2013) showed that a grid plate covering the central slot can achieve the best VIV suppression of twin-box girders. Although the results showed that countermeasures like increasing structural damping ratio, installing grid plates, wind barriers and guide vanes have positive effect on reducing VIV responses of long-span bridges with twin-box girders, the effectiveness of these method are dependent on the geometrical configuration of bridge sections (Sarpkaya, 2004) which may closely related to the slot width ratio. Therefore, the further research work on identifying the most effective measure for suppressing the VIV of twin-box girder bridges with various slot width ratios is required.

The purpose of the present study is to comprehensively investigate the VIV performance of a long-span twin-box girder bridge with various slot width ratios and to gain a systematic evaluation on the effectiveness of various VIV countermeasures through a series of wind tunnel tests. Firstly, the sectional models with two geometrical scaled ratios of 1:60 and 1:30 and six representative slot width ratios varying from 0% to 100% were selected and designed. Then, the VIV responses of two scaled models in heaving and torsional degree of freedom (DOF) under three wind attack angles were systematically tested and analyzed. Finally, by selecting two slot width ratios (i.e. 20% and 60%) as the control cases, four countermeasures (i.e. increase of structural damping ratio, installation of grid plates, adjustable wind barriers and guide vanes) were employed to evaluate their effectiveness with the aim of identifying available VIV countermeasure strategies, based on the peak amplitudes of the bridge by two important criteria (e.g. the displacement requirement of Sperling indicator and Wind-resistance specification of China). In summary, the present study could potentially contribute to further understanding the VIV performance of twin-box girders bridges and allow the more reasonable selection of countermeasures.

2. Wind tunnel experiment set-up of VIV

A series of wind tunnel tests were performed on elastically suspended models that capture the aerodynamic characteristic of the stiffening girder of the Xihoumen suspension bridge in China. The VIV tests involving two models (i.e. 1:60

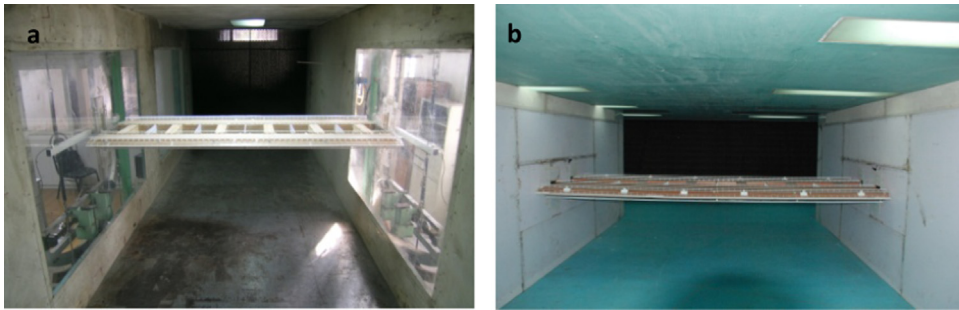


Fig. 1. Sectional diagrams of two scale models: (a) 1:30 scale ratio, and (b) 1:60 scale ratio.

and 1:30 geometric scale ratios), with low and high Reynolds numbers, were conducted in the TJ-1 and TJ-3 Boundary Layer Wind Tunnel at Tongji University (Fig. 1), respectively. The suspension system consists of 8 springs and supporting frames, while 4 laser displacement transducers were used for measuring the vertical and torsional response of two sectional models to vortex-shedding excitation. Two rigid and streamlined temporary end walls were also constructed for TJ-3 BLWT to accommodate the 1:30 sectional model (Fig. 1(b)). As shown in Fig. 2, the sectional model with 1:30 scale ratio is 0.5 m wide for each single girder ($B_s/2$) and 0.117 m high (H), resulting in an aspect ratio $8.55B_s/H$. Table 1 listed the structural parameters including geometric dimensions, mass characteristics, fundamental frequencies and structural damping of the 1:30 scale model. In order to comprehensively investigate the influence of slot width ratio (i.e. D to B_s) on the VIV performance in a suspension bridge with twin-box girders, six representative test configurations were designed for both two scale models, with a slot width ratio ranging from 0%, 20%, 40%, 60%, 80% and 100%. Two groups of mass moment of inertia (MMI) were used with consideration of the change of structural stiffness due to the fact that MMI becomes larger with the increase of slot width ratio. In the experiments, we adopted adjustable additional mass on the section models to make the equivalent mass consistent under different slot width ratios. The first group consisted three relatively small slot width ratios (i.e. 0%, 20% and 40%) by keeping the value of MMI of the 20% slot width ratio ($4.723 \text{ kg m}^2/\text{m}$), while three relatively large ratios (i.e. 60%, 80% and 100%) were assumed to the value of MMI of the 80% slot width ratio ($9.642 \text{ kg m}^2/\text{m}$) in the second group. Totally, there were 18 testing cases under three wind attack angles (α) of -3° , 0° and $+3^\circ$ for both models, as shown in Table 2.

3. VIV responses of twin-box girders with two scale models

According to the results of previous studies (Larsen et al., 2008; Yang et al., 2015), the 1:60 geometrical scale ratio, as a common model scale from 1:50 to 1:80, may not be able to accurately simulate the aerodynamic characteristic of the complex girder deck. Thus, the low and high Reynolds number experiment with the scale ratio of 1:60 and 1:30 were conducted to explore the results of VIV performance of twin-box girder with various slot width ratios, respectively. The Strouhal Number and Reynolds Number of the 1:60 model with the 20% slot ratio at the VIV condition are about 0.127 and 18070, respectively. The Strouhal Number and Reynolds Number of the 1:30 model with the 20% slot ratio at the VIV condition are about 0.133 and 34240, respectively.

3.1. Sectional model with 1:60 scale ratio

3.1.1. Heaving VIV responses

VIV responses in heaving DOF of six slot width ratios under three wind attack angles of -3° , 0° and $+3^\circ$ are described in Fig. 3(a)–(c), respectively. The vertical response is normalized by that of the actual bridge (relative to the height of the main girder). The normalized peak amplitude of the 60% slot width ratio with the value of 0.094 is the largest, followed by the 40% and 80% slot width ratio under the $\alpha = +3^\circ$, while the amplitude of the 0% slot width ratio is the smallest among six slot

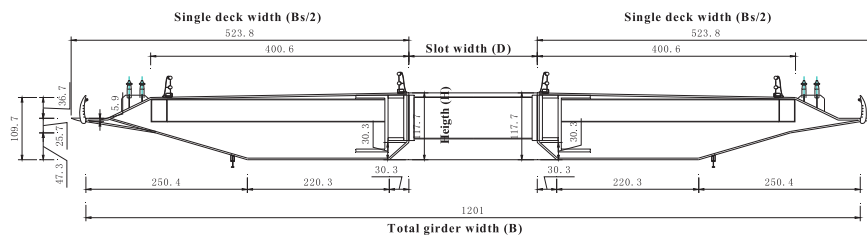


Fig. 2. Sectional model diagrams of 1:30 scale ratio (unit: mm).

Table 1

Structural parameters of sectional model with 1:30 scale ratio.

Properties		Slot width ratio					
		0%	20%	40%	60%	80%	100%
Geometric scale	Length L/m	3.6	3.6	3.6	3.6	3.6	3.6
	Width B/m	1.0	1.2	1.4	1.6	1.8	2.0
	Height H/m	0.117	0.117	0.117	0.117	0.117	0.117
Equivalent mass	Mass/unit length m /kg/m	29.326	29.326	29.326	29.326	29.326	29.326
	Mass moment of inertia/unit length I_m / kg m ² /m	4.723	4.723	4.723	9.642	9.642	9.642
Frequency	Vertical f_h /Hz	4.86	4.86	4.86	4.86	4.86	4.86
	Torsional f_t /Hz	6.99	6.99	6.99	6.99	6.99	6.99
Damping ratio	Vertical ξ_h /‰	2.5	2.5	2.5	2.5	2.5	2.5
	Torsional ξ_t /‰	2.5	2.5	2.5	2.5	2.5	2.5

Table 2

Testing cases of VIV performance.

Scale ratio	Name	Slot width ratio	Slot width D (m)	Deck width B_s (m)	Total width B (m)	Wind attack angle (°)
1:60	D00-1	0%	0.0	1.0	1.0	0, ± 3
	D20-1	20%	0.2	1.0	1.2	0, ± 3
	D40-1	40%	0.4	1.0	1.4	0, ± 3
	D60-1	60%	0.6	1.0	1.6	0, ± 3
	D80-1	80%	0.8	1.0	1.8	0, ± 3
	D100-1	100%	1.0	1.0	2.0	0, ± 3
1:30	D00-2	0%	0.0	0.5	0.5	0, ± 3
	D20-2	20%	0.1	0.5	0.6	0, ± 3
	D40-2	40%	0.2	0.5	0.7	0, ± 3
	D60-2	60%	0.3	0.5	0.8	0, ± 3
	D80-2	80%	0.4	0.5	0.9	0, ± 3
	D100-2	100%	0.5	0.5	1.0	0, ± 3

width ratios. The results show that the single box girder with the 0% slot width ratio produces the best VIV performance in comparison to other slot width ratios. It is interesting that there are two lock-in regions for two larger slot width ratios of 80% and 100%, in which the second lock-in region ranging from about the reduced wind velocity ($U/f_n B$) of 0.54–0.84 has the larger amplitudes and higher wind velocities. As shown in the Fig. 3(d), the most unfavorable behavior is given by the wind attack angle of 0° when the slot width ratio is smaller than 40%, and then change to the +3° angle after the 40% slot width ratio. In addition, all of the peak amplitudes with −3° wind attack angle are the smallest among three angles. It is noted that heaving VIV responses present striking upward trend with an increase in slot width ratio before the 40% and then continuously decrease after reaching 60% slot width ratio.

3.1.2. Torsional VIV responses

Different from the heaving VIV responses of twin-box girders, the peak amplitude of the 80% slot width ratio in torsional VIV responses is the largest with the value of 0.932 at the reduced wind velocity ($U/f_t B$) of 0.52, which is larger than that of the 100% and 40% slot width ratio (0.852 and 0.850). More than one lock-in region for three larger slot width ratios of 60%, 80% and 100% are emerged, in which all the amplitudes in the first lock-in region with the $U/f_t B$ of about 0.5 are much larger than those in the second one. In this case, the maximum torsional displacements are taken place at the lower wind velocity by compared with the heaving VIV. Besides, the VIV phenomenon cannot be observed for the 0% slot width ratio and torsional VIV performance with the 60% slot width ratio is inferior to other slot width ratios. Similarly, the −3° wind attack angle is the most favorable condition for torsional VIV responses, as presented in Fig. 4.

Above all, both of heaving and torsional VIV responses show a great increment after slotting and the VIV performance for a twin-box girder is sensitive to the slot width ratio. Specifically, although the 60% slot width ratio has the maximum amplitude in heaving VIV responses, its torsional amplitude is the smallest among five slot width ratios. The 20% slot width ratio has the best VIV performance in terms of the heaving and torsional VIV responses expect for the single box girder (0% slot width ratio).

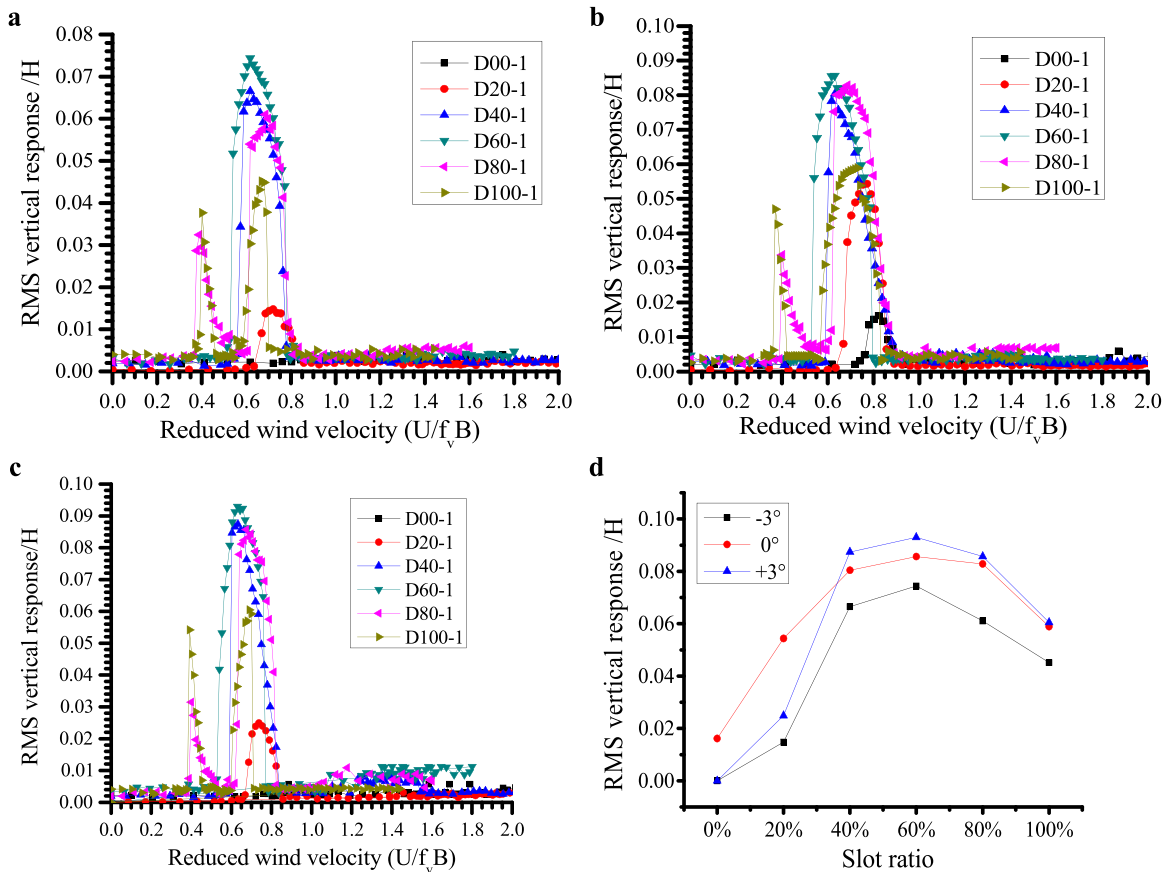


Fig. 3. Heaving VIV responses of the 1:60 scale ratio. (a) $\alpha = -3^\circ$, (b) $\alpha = +0^\circ$, (c) $\alpha = +3^\circ$, and (d) peak amplitudes.

3.2. Sectional models with 1:30 scale ratio

3.2.1. Heaving VIV responses

As shown in the Fig. 5, the normalized peak amplitude of six slot width ratios in heaving is 0.105 for the 80% slot width ratio (i.e. at the $U/f_v B$ of 0.674), which is larger than that for 1:60 scale ratio with the value of 0.940. Nevertheless, the heaving VIV responses of the 1:30 scale ratio for three small slot width ratios of 0%, 20% and 40% are slight smaller than those corresponding responses of the scale ratio of 1:60. Under the most unfavorable wind attack angles of $+3^\circ$, the magnitude of vertical displacements rapidly increases, reaches its peak at the 80% slot width ratio, and then dramatically decreases. It is interesting that the 0% slot width ratio also has not emerged the VIV phenomenon and all the torsional responses under -3° wind attack angle are also the smallest among three angles. In addition, the relatively large amplitudes and high wind velocities are also found in the second lock-in regions with a slot width ratio ranging from 80% to 100%.

3.2.2. Torsional VIV responses

Both the 0% and 20% slot width ratio in 1:30 scale ratio have not the torsional VIV phenomenon, which is different from the 20% slot width ratio in the 1:60 scale ratio, as illustrated in Fig. 6. A obvious lock-in region could be found in the torsional responses of 1:30 scale ratio, and only the peak amplitudes of the 40% slot width ratio under three wind angles appear at a higher reduced wind velocity about 0.8. The maximum torsional displacements of the 1:60 and 1:30 scale models under the wind attack angles of $+3^\circ$ are 0.991 and 0.932, respectively. Besides, the peak responses from the 40–80% slot width ratio for the 1:30 scale ratio are close to those for the 1:60 scale ratio. It can be seen clearly that, the amplitudes of the 100% slot width ratio are the largest among six slot width ratios, regardless of three wind attack angles. However, the torsional response of the 60% slot width ratio is much smaller than other three larger slot width ratios. Most importantly, the reduced wind velocities in lock-in region corresponding to the peak amplitude are about 0.5, which indicates that the torsional VIV phenomenon could be happened even in low wind velocity.

To sum up, the VIV performance of the single box girder with the 0% slot width ratio is superior to the twin-box girder with other slot width ratios. The heaving VIV responses of twin-box girders firstly increase, reaches to its peak, and then gradually decrease with the increase of slot width ratio. The heaving responses of the 60% slot width ratio are too large to be suppressed by using available countermeasures. In particular, the difference of the torsional responses of the 20% slot width

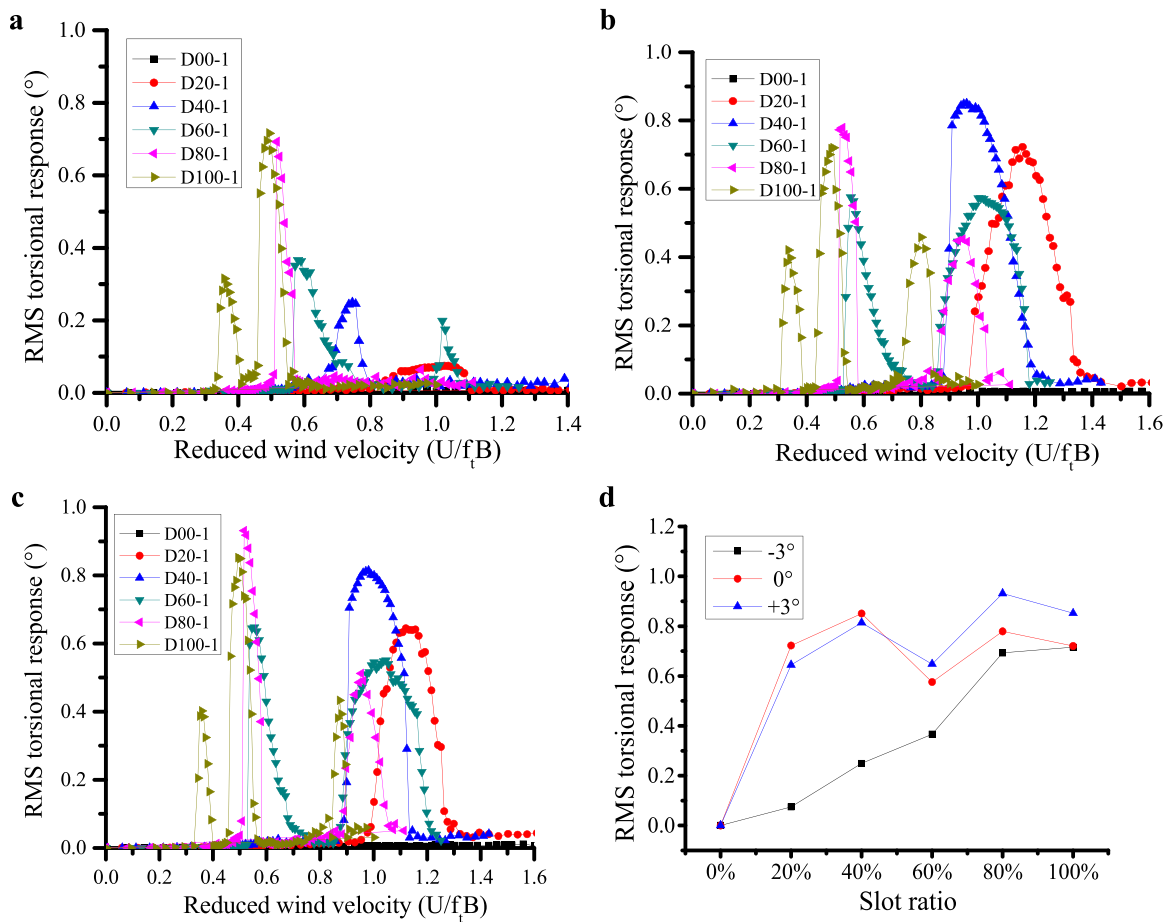


Fig. 4. Torsional VIV responses of the 1:60 scale ratio. (a) $\alpha = -3^\circ$, (b) $\alpha = 0^\circ$, (c) $\alpha = +3^\circ$, and (d) peak amplitudes.

ratio between two scale models is obvious. Meanwhile, the slot width ratio of twin box girder is about 20% of the Xihoumen Bridge and 25% of Shanghai Yangtze River Bridge (Yang et al., 2014), which indicates that the 20% is a popular slot width ratio in long-span bridges with twin-box girders. Therefore, the 20% and 60% slot width ratio are selected as the control when evaluating the control effects of VIV countermeasures.

4. Countermeasures of vortex-induced vibration

It is important to ensure sufficiently high test Reynolds numbers with a large geometrical scale ratio to avoid thick boundary layers growing on the testing model and capture the aerodynamic characteristic of countermeasures (Larsen et al., 2008; Yang et al., 2015). With the consideration of geometrical, mass, stiffness and Reynolds number, the sectional model with a 1:30 geometrical scale ratio was selected to further investigate the VIV control measures. Using the 20% slot width ratio as the reference, the VIV tests of four countermeasures were conducted in smooth flow at Tongji University's TJ-3 Boundary Layer Wind Tunnel to evaluate the effectiveness of increasing structural damping ratio, installing grid plates, wind barriers and guide vanes in VIV control. The results of previous studies (Ge et al., 2011; Yang et al., 2014) have indicated that increasing structural damping ratio and installing grid plates have excellent capability of mitigating the heaving VIV. In this study, the control effectiveness of increasing damping ratio and installing grid plates on the suppression of heaving VIV for the twin-box girder bridge with 60% slot width ratio were also studied. The detail testing cases are listed in Table 3.

4.1. Increasing structural damping ratio

The original structural damping ratios of sectional models was around 0.25%. In this study, the damping ratio was increased to 0.5% through inserting lead retainers into the springs of elastically suspended models. The comparisons of the maximum VIV responses under three wind angles for the 20% and 60% slot width ratios with two damping ratios are illustrated in Fig. 7.

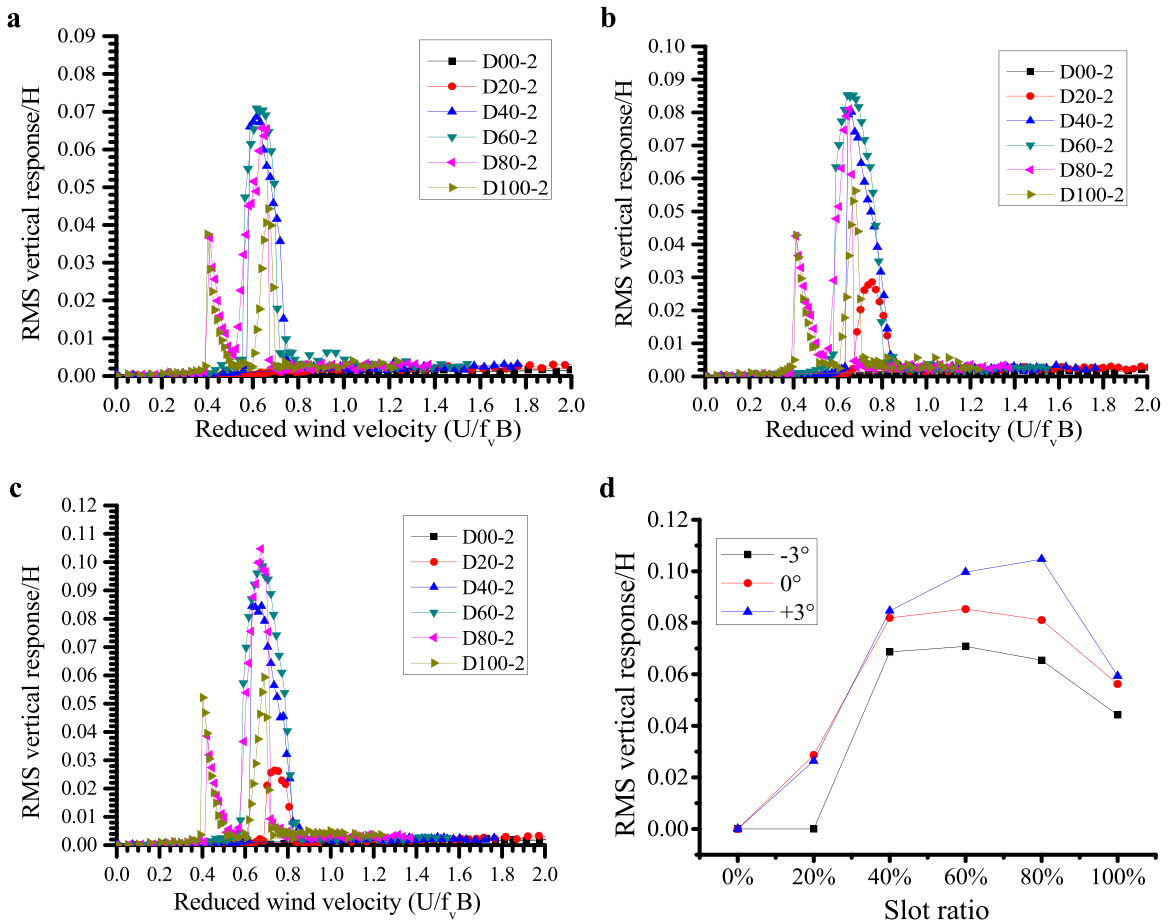


Fig. 5. Heaving VIV responses of the 1:30 scale ratio. (a) $\alpha = -3^\circ$, (b) $\alpha = +0^\circ$, (c) $\alpha = +3^\circ$, and (d) peak amplitudes.

Fig. 7 indicates that all these VIV responses become smaller as the increase of structural damping ratio, especially for the heaving response of the 20% slot width ratio. Specifically, both the normalized heaving and torsional amplitude of the 60% slot width ratio decreased by -26% and -41% , respectively. However, the heaving amplitude of the 20% slot width ratio striking decrease from 0.0286 (at the reduced wind velocity of 0.75) to 0.0036 with the change of about 87%. When the structural damping ratio was increased from 0.25% to 0.5%, the VIV responses of the 60% slot width ratio decreased by about 40%, while the VIV responses of the 20% slot width ratio decreased by about 87%. The different control effects of VIV responses for increasing damping ratio probably are contributed to the aerodynamic shape of twin box girder, since the positive aerodynamic damping force and flow characteristics which mainly affect the maximum amplitude of VIV are sensitive to the aerodynamic shape. Therefore, regardless of heaving and torsional VIV, the increase of structural damping ratio is useful for mitigating VIV responses and the effectiveness for the 20% slot width ratio are more obvious than that of the 60% slot width ratio.

4.2. Installing grid plates

As a new effective VIV countermeasure, grid plates covering the central vent are used to block the airflow and interference with the formation of large scale vortices at the central slot, since strong vortices generated at the central slot is one of the critical factor for inducing the large amplitude of vortex-induced vibration (Laima et al., 2013; Yang et al., 2014). Grid plates with five different grid porosities varying from 0% to 70% are proposed to control the VIV of twin-box girders and their control effectiveness are compared with the original section, where two examples of the 0% and 70% grid porosities are simply introduced in the Fig. 8(b) and (c).

Fig. 9 presents the comparison of maximum heaving and torsional responses of twin-box girders with two slot width ratios. As for the 20% slot width ratio, the grid plates with all the five grid porosities have great influence on the suppression of heaving VIV since the amplitudes directly reduce from 0.0286 to about 0.0030 with the change of about 89%. Although the amplitude of torsional VIV for the original section is very small with the value of 0.0538, the control effectiveness of grid plates is obvious only when the grid porosity becomes smaller than 30%. With respect to the 60% slot width ratio, the effectiveness of grid plates on VIV mitigation gradually improves with the decrease of grid porosity, in which the change of

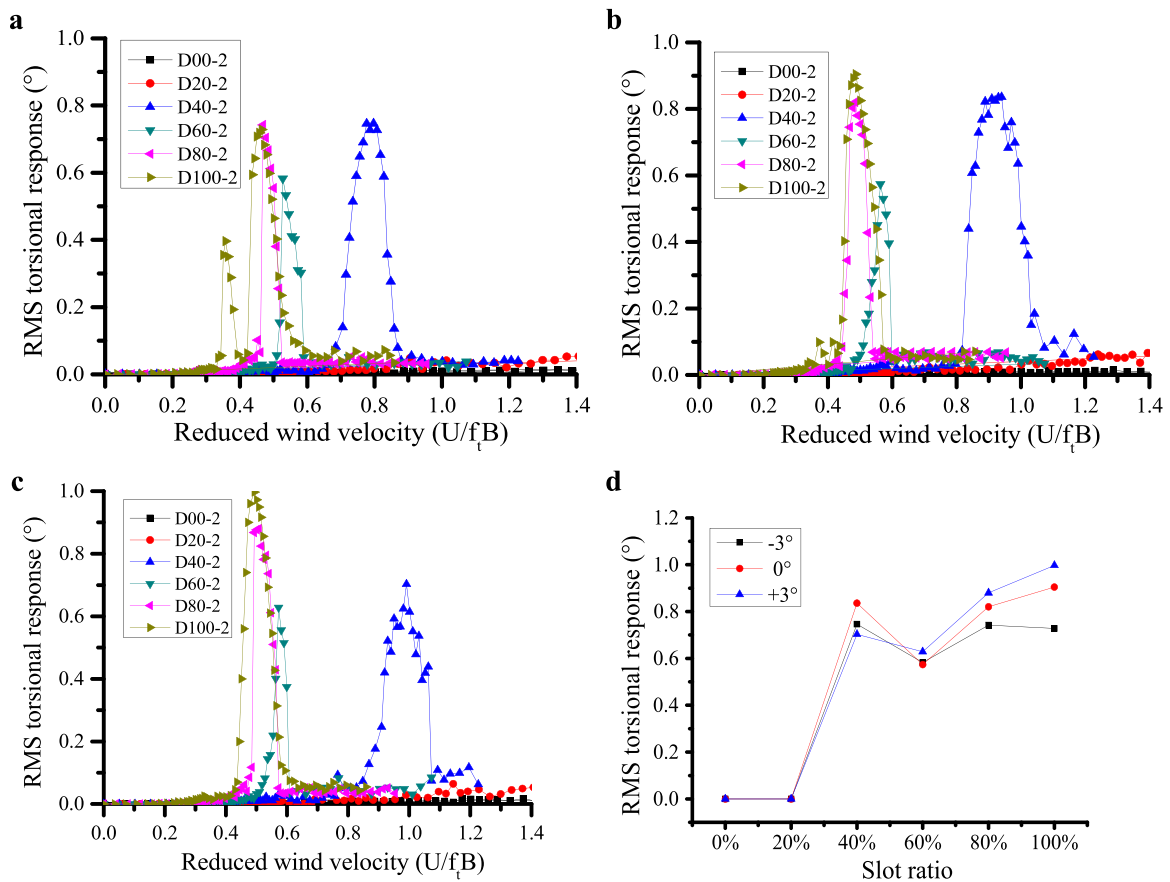


Fig. 6. Torsional VIV responses of the 1:30 scale ratio. (a) $\alpha = -3^\circ$, (b) $\alpha = 0^\circ$, (c) $\alpha = +3^\circ$, and (d) peak amplitudes.

Table 3

Testing cases of VIV countermeasures.

Slot width ratio	Name	Counter measures	Parameters	Wind attack angle ($^\circ$)
20%	D20-2-0	No	–	$0, \pm 3$
	D20-2-1	Increasing damping ratio	0.5%	$0, \pm 3$
	D20-2-2	Installing grid plates	70%	$0, \pm 3$
	D20-2-3		50%	$0, \pm 3$
	D20-2-4		30%	$0, \pm 3$
	D20-2-5		0%	$0, \pm 3$
	D20-2-6	Installing wind barrier	Vertical	$0, \pm 3$
	D20-2-7		Horizontal	$0, \pm 3$
	D20-2-8		Internal	$0, \pm 3$
	D20-2-9		External	$0, \pm 3$
60%	D60-2-0	No	Control	$0, \pm 3$
	D60-2-1	Increasing damping ratio	0.5%	$0, \pm 3$
	D60-2-2	Installing grid plates	70%	$0, \pm 3$
	D60-2-3		50%	$0, \pm 3$
	D60-2-4		30%	$0, \pm 3$
	D60-2-5		0%	$0, \pm 3$

the 30% and 70% grid porosity are about -70% and -31% , respectively. Moreover, the oscillation amplitudes of the 0% grid porosity are close to zero, and the vortex-induced vibration is completely suppressed. It is worth noting that the torsional response of the 60% slot width ratio whereas arise from 0.393 (i.e. at the lower reduced wind velocity of 0.57) to 0.399 (i.e. at the higher wind velocity of 0.81) after installing grid plates with 50% grid porosity. Fortunately, not only the peak amplitudes of torsional VIV of the 60% slot width ratio slight decline, but also the reduced wind velocity of lock-in region become

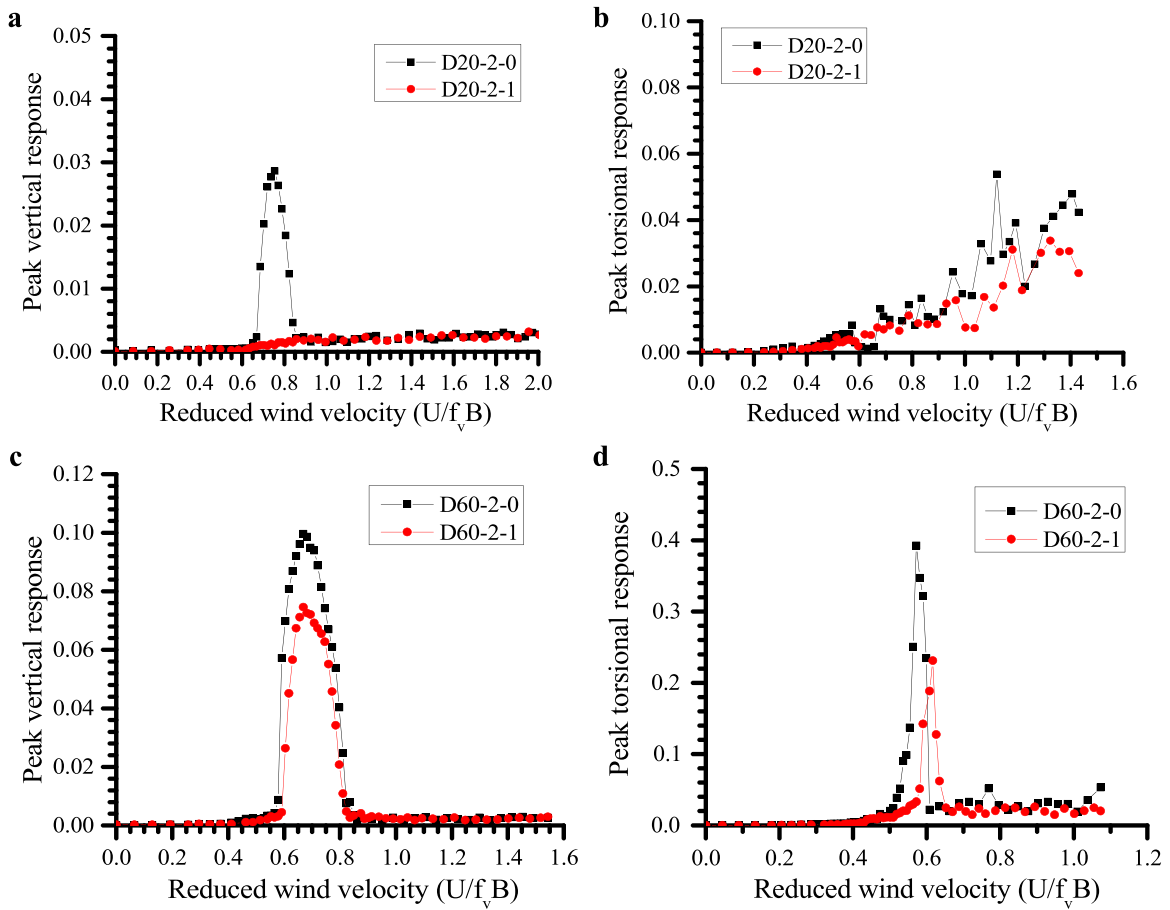


Fig. 7. Comparison of increasing damping ratio. (a) heaving responses of 20% slot width ratio, (b) torsional responses of 20% slot width ratio, (c) heaving responses of 60% slot width ratio, and (d) torsional responses of 60% slot width ratio.

higher to around 0.81, which is benefited to reduce the probability of VIV in low wind velocity. Similar to the heaving VIV, the grid plates with 0% grid porosity has the best control effectiveness on torsional VIV. Therefore, the grid plate is an excellent countermeasure for suppressing the heaving VIV, particularly with 20% slot width ratio. Although the grid plate has limited effect on reducing torsional VIV for the 60% slot width ratio, it makes the appearance of the peak amplitude only occur at a higher wind velocity.

4.3. Installing wind barriers

In order to improve the serviceability of long-span bridges and guarantee the safe driving of vehicles, adjustable wind barriers were proposed to be installed on the top of the guard rails on the two sides of the girder deck. Through changing the flow pattern on the surface of the girder deck, the application of adjustable wind barriers could affect the flow characteristic and vortices formation (Ge et al., 2011; Yang et al., 2014). As a consequence, the adjustable wind barriers referring to full scale in Fig. 8(a) were also designed to mitigate the VIV of twin-box girders with 20% slot width ratio, and the control effectiveness of wind barriers in both horizontal and vertical positions were evaluated, as shown in Fig. 10.

The results show that there is no obvious VIV detected with wind barriers added in vertical or horizontal position. The maximum torsional amplitude is increased from 0.0538 to 0.0582 and 0.0713 with barriers laid down in vertical and horizontal position, respectively. Besides, the maximum amplitude of heaving VIV is always reduced from 0.0286 to about 0.004 for two positions of wind barriers. Therefore, these two types of wind barriers have great potential in suppressing the heaving VIV but have adverse influence in reducing torsional responses, in which the wind barriers in vertical position is less effective than that in horizontal position.

4.4. Installing guide vanes

Guide vane can smooth air flow around the corner of girder section and is a feasible VIV control solution, and the effectiveness of guide vane is closely related to the position and dimension of guide vane (Larsen et al., 2008; Li and Ge,

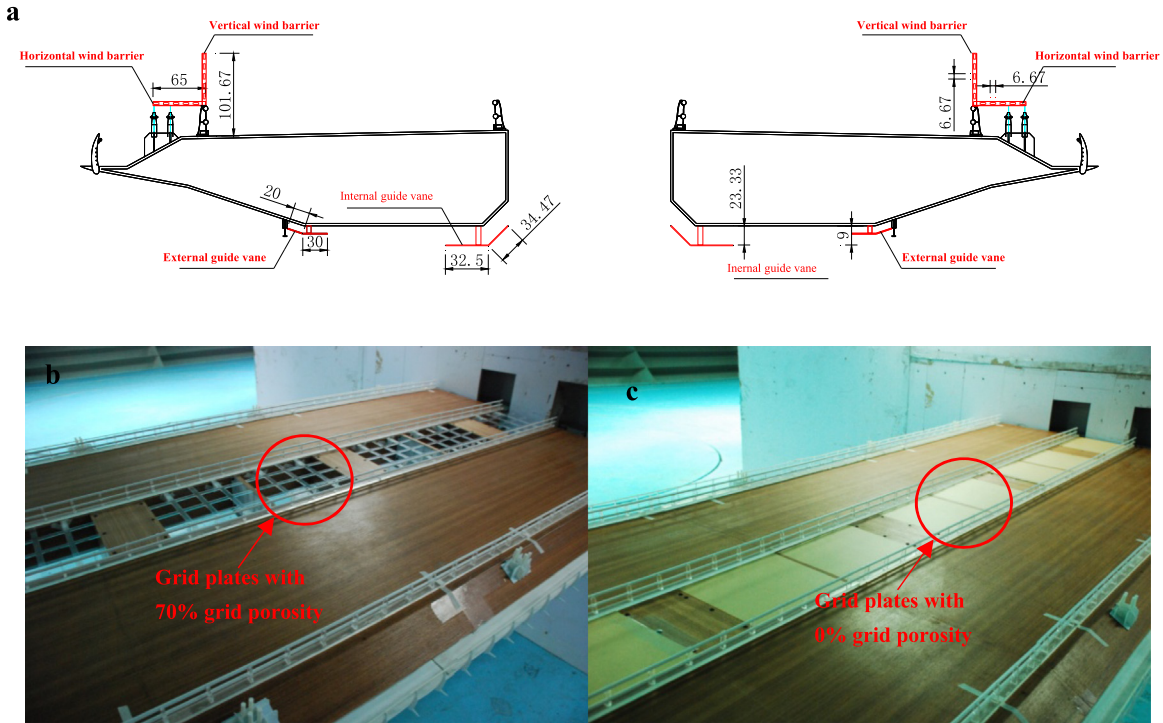


Fig. 8. Scheme of VIV countermeasures in 1:30 model. (a) guide vanes and wind barriers (unit: mm), (b) grid plates with 70% grid porosity, and (c) grid plates with 0% grid porosity.

2008). Two locations of guide vanes are used to mitigate the oscillations induced by vortex shedding. The “internal guide vanes” is installed at the bottom slab near central slot, while the “external guide vanes” is installed near the outside of girder deck (Fig. 7(c)). The VIV responses of twin-box girders with these two types of guide vanes were shown in Fig. 11.

It can be seen that guide vanes are detrimental to reduce the amplitudes of both heaving and torsional VIV of the 20% slot width ratio. The effectiveness of guide vanes are different from that in literature (Ge et al., 2011), which shows the control effects of guide vanes are associated with the slot width ratio. In this study, the peak amplitude of heaving VIV increase from 0.0286 to 0.0364 and 0.0339 for the internal and external guide vanes with the growth rate of about 27% and 18%, respectively. In addition, the growth rate of torsional peak amplitude of internal guide vanes also reaches 26%, while the external guide vanes are benefited to the torsional VIV of twin-box girder to some extent. Therefore, these two types of guide vanes cannot effectively suppress the VIV responses of the twin-box girder with the 20% slot width ratio. The implication of these results is that the design of internal guide vanes should be further improved and investigated in term of dimensions and angles of guide vanes.

4.5. Evaluation of four countermeasures

4.5.1. Two evaluation indicators

In order to meet the requirement of the safety and serviceability of bridge structures, there are some different stipulates on the amplitude limitation of VIV responses. In this study, two common stipulates are used to comprehensively evaluate the feasibility of the four countermeasures on improving the VIV performance of the suspension bridge with twin-box girders.

Firstly, according to the Chinese specification JTG/T D60-01-2004, ‘Wind-resistant design specification for bridges’ (given by the Chinese Ministry of Communications and abbreviated the JTG/T), the amplitude limitation of the heaving and torsional VIV of the real bridge with 20% slot width ratio are defined as follows (where f_h , B , f_t are listed in Table 1):

$$0.04/f_h = 0.04/0.324 = 0.124\text{m} \quad (1)$$

$$4.56/(Bf_t) = 4.56/(36 \times 0.233)\text{m} = 0.544^\circ \quad (2)$$

For the actual bridge with the 60% slot width ratio, the heaving and torsional limitation of VIV responses are 0.124 m and 0.408° , respectively.

Secondly, as an important indicator of the driving comfortable of bridge structures, Sperling indicator can also be adopted to conduct the evaluation of VIV performance due to the single frequency of VIV phenomenon (Garg and Dukkipati, 1984). The mathematical expression of Sperling indicator W has the following form:

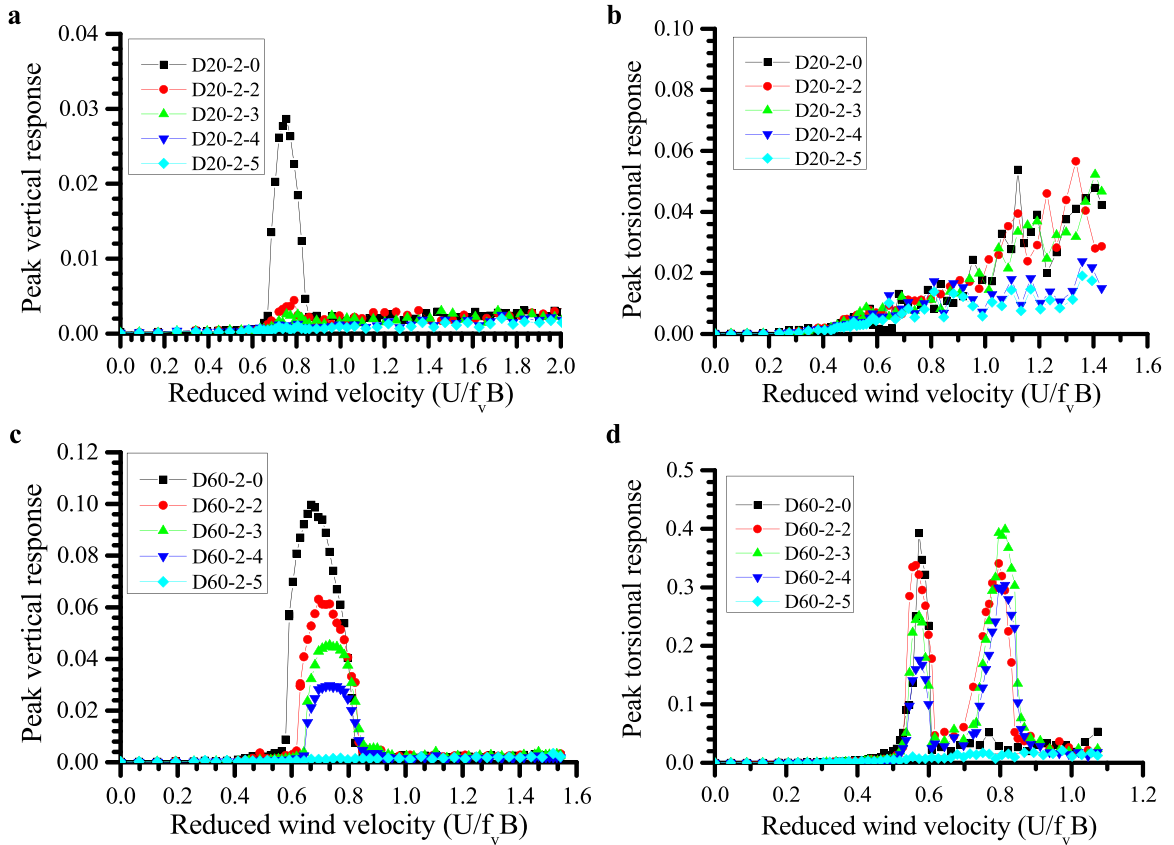


Fig. 9. Comparison of installing grid plates. (a) heaving responses of 20% slot width ratio, (b) torsional responses of 20% slot width ratio, (c) heaving responses of 60% slot width ratio, and (d) torsional responses of 60% slot width ratio.

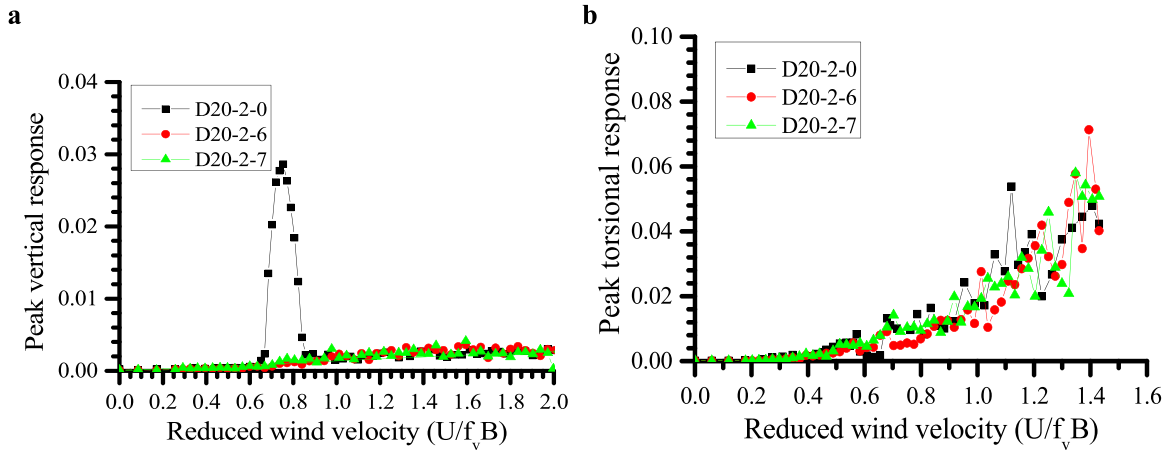


Fig. 10. Comparison of installing wind barriers. (a) heaving responses of 20% slot width ratio, (b) torsional responses of 20% slot width ratio.

$$W = 2.7^{10} \sqrt{z_0^3 f^5 F(f)} = 0.896^{10} \sqrt{a^3 F(f)} / f \quad (3)$$

where z_0 is the peak amplitude. f is the structural frequency. a is the acceleration and can be calculated as follows: $a = z_0 (2\pi f)^2$. $F(f)$ is the weighting coefficient related to the vibration frequency. It should be noted that the frequency f is assumed to 0.5 and then $F(f) = 0.325 f^2 = 0.325 \times 0.5^2 = 0.0813$.

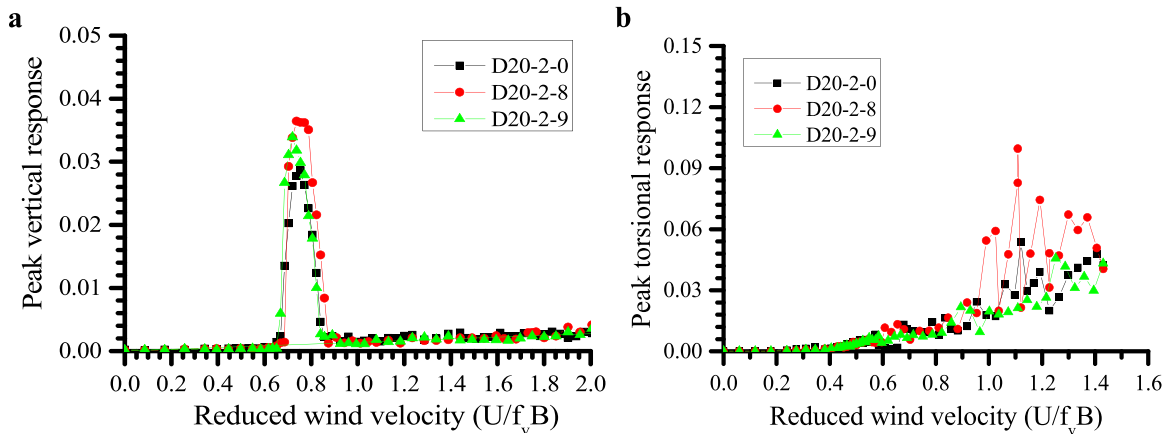


Fig. 11. Comparison of installing guide vanes. (a) heaving responses of 20% slot width ratio, (b) torsional responses of 20% slot width ratio.

Table 4
Sperling indicator W.

W	Vibration sensitivity
1	Just noticeable
2	Clearly noticeable
2.5	More pronounced but not unpleasant
3	Strong, irregular, but still tolerable
3.25	Very irregular
3.5	Extremely irregular, unpleasant, annoying, prolonged exposure intolerable
4	Extremely unpleasant; prolonged exposure harmful

The Sperling indicator W is used to reflect the driving comfort according to the definition in Table 4, and the value of $W=3$ is regarded as the standard in this study. As a result, the amplitude limitation of the torsional VIV for the 20% and 60% slot width ratio are 1.184° and 0.888° , respectively, with a same heaving amplitude limitation of 0.215 m. It is found that the

Table 5
Comprehensive evaluation of four countermeasures in actual bridge.

Cases	Evaluation of heaving VIV performance				Cases	Evaluation of torsional VIV performance			
	Peak amplitude (m)	JTG/T 0.1235 m	$W=0.2146$ m	Change percentage		Peak amplitude ($^\circ$)	JTG/T 0.5436°	$W=1.1835^\circ$	Change percentage
D20-2-0	0.1005	satisfied	satisfied	0%	D20-2-0	0.054	satisfied	satisfied	0%
D20-2-1	0.0126	satisfied	satisfied	–87.5%	D20-2-1	0.036	satisfied	satisfied	–33.3%
D20-2-2	0.0156	satisfied	satisfied	–84.5%	D20-2-2	0.057	satisfied	satisfied	5.5%
D20-2-3	0.0129	satisfied	satisfied	–87.2%	D20-2-3	0.052	satisfied	satisfied	–3.7%
D20-2-4	0.0102	satisfied	satisfied	–89.9%	D20-2-4	0.024	satisfied	satisfied	–55.6%
D20-2-5	0.0088	satisfied	satisfied	–91.2%	D20-2-5	0.019	satisfied	satisfied	–64.8%
D20-2-6	0.0133	satisfied	satisfied	–86.8%	D20-2-6	0.071	satisfied	satisfied	31.5%
D20-2-7	0.0144	satisfied	satisfied	–85.7%	D20-2-7	0.0058	satisfied	satisfied	7.4%
D20-2-8	0.1328	Not satisfied	satisfied	32.1%	D20-2-8	0.111	satisfied	satisfied	105.5%
D20-2-9	0.1278	Not satisfied	satisfied	27.2%	D20-2-9	0.050	satisfied	satisfied	–7.5%
Cases	Evaluation of heaving VIV performance				Cases	Evaluation of torsional VIV performance			
	Peak amplitude (m)	JTG/T 0.1235 m	$W=0.2146$ m	Change percentage		Peak amplitude ($^\circ$)	JTG/T 0.4077°	$W=0.8876^\circ$	Change percentage
D60-2-0	0.3498	Not satisfied	Not satisfied	0%	D60-2-0	0.3926	satisfied	satisfied	0%
D60-2-1	0.2619	satisfied	Not satisfied	–25.1%	D60-2-1	0.2313	satisfied	satisfied	–41.1%
D60-2-2	0.2217	Not satisfied	Not satisfied	–36.6%	D60-2-2	0.3567	satisfied	satisfied	–9.1%
D60-2-3	0.1710	satisfied	satisfied	–51.1%	D60-2-3	0.3991	satisfied	satisfied	+1.7%
D60-2-4	0.1195	satisfied	satisfied	–65.8%	D60-2-4	0.3041	satisfied	satisfied	–22.5%
D60-2-5	0.0011	satisfied	satisfied	–99.7%	D60-2-5	0.044	satisfied	satisfied	–88.8%

amplitude limitation of JTG/T is close to half of Sperling indicator, and thus the requirements of JTG/T are more stringent than that of Sperling indicator.

4.5.2. Comprehensive evaluation

Based on the limitations of JTG/T and Sperling indicator, the maximum VIV amplitudes of a twin-box girder bridge with and without countermeasures are evaluated, as shown in Table 5. The measured displacements are converted to that of the actual bridge, in which the positive percentage of change represents the increase in amplitude whereas the negative percentage means the decrease in amplitude.

Table 5 indicates that most of the VIV responses for the 20% slot width ratio girder are satisfied with displacement requirement of two stipulates, except for two cases of D20-2-8 and D20-2-9. For the heaving VIV responses of the bridge with two locations of guide vanes, both of the peak amplitudes are smaller than the requirement of Sperling indicator but surpass the requirement of JTG/T. Although the change of three cases including D20-2-6, D20-2-7 and D20-2-8 are positive percentages, the peak amplitudes of the three cases are much smaller than those of two stipulates. In contrast, only two cases of D60-2-4 and D60-2-5 are satisfied with the heaving requirement of JTG/T and another case of the D60-2-3 with the change of -51.5% is also satisfied with the requirement of Sperling indicator. In other word, the VIV performance of the bridge with the 60% slot width ratio by increasing damping ratio and installing grid plates does not accord with the heaving VIV limitation of JTG/T. In addition, the case of the grid plates with the 50% grid porosity has the positive change of $+7.1\%$, whereas is smaller than the torsional VIV limitation of JTG/T. Besides, the torsional peak amplitudes of other five cases are also satisfied with the requirements of two stipulates. Therefore, heaving VIV problem of the long-span bridge with twin-box girder should be paid more attention and the implementation of available countermeasures to suppress the heaving responses is recommended. Only the 60% slot width ratio bridge with the 0% and 30% grid porosities have completely met the requirements of two stipulates after applying grid plates, while for the bridge with the 20% slot width ratio, the installation of guide vanes will not help the bridge to meet the requirement of JTG/T.

5. Conclusion

In present study, vortex-induced vibration performance of a twin-box girder bridge with six slot width ratios was systematically studied and then the effectiveness of four VIV countermeasures were compared and evaluated with two stipulates (i.e. Sperling indicator and Wind-resistance specification of China). The major conclusions are as follows:

- The VIV performance of the single box girder with 0% slot ratio is superior to those with central slot. The torsional responses of twin-box girders generally increase with the increase of slot ratio, except for the 60% slot ratio. It is worth noting that a significant difference in torsional responses between two scale models is found at the slot width ratio of 20%.
- The heaving VIV responses of twin-box girders firstly increase with the slot ratio, reach to its peak when the slot ratio is 60% for 1:60 geometrical scale model and 80% for 1:30 scale model, and then gradually decrease as slot ratio increases.
- The effects of increasing damping ratio on the mitigation of VIV responses of twin box girders show that the potential application of some mitigation measures, for example, dampers, while installing guide vanes may have adverse effects. The installation of grid plates and wind barriers can significantly reduce the heaving VIV amplitudes of twin-box girders with 20% slot ratio. The control effect of grid plates on heaving VIV gradually diminish with the increase of the grid porosity, regardless of the 20% and 60% slot ratio.
- The torsional VIV displacements of the twin-box girder bridge are satisfied with the requirements of two stipulates even without countermeasures. The heaving VIV displacements for the 20% slot ratio bridge after installing guide vanes, and heaving displacements for the 60% slot ratio bridge after increasing damping ratio and installing grid plates with the 70% grid porosity not meet the requirement of JTG/T D60-01-2004.

Although the installation of grid plates with the 0% grid porosity has the best control effectiveness on VIV performance of the twin-box girder bridge, installing grid plates with 0% grid porosity has completely changed the aerodynamic behavior, which may lead to instability problems, and therefore needs to be further studied. The influence of the grid plates of different porosities and other countermeasures on the flutter performance of twin box girders should also be investigated when these VIV mitigation measures are potential applied to the bridges in practice. In addition, further research work is required to investigate the fundamental mechanisms of a series of VIV countermeasures and the influence of Reynolds number on the VIV performance of twin-box girder.

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